

Product Increment

Project Title: Enhancement and verification of Python-based software for image-guided surgery

Increment Name: “Sprint 2 Calibration & Reconstruction Pipeline Increment for Ctrl+Incision”

Core user journey supported end-to-end flow:

1. System Setup & Initialization
 - a. User installs required dependencies stated in README.md (Python, PLUS ToolKit, etc.).
 - b. The environment variable PLUS_TOOLKIT_PATH is configured.
 - c. User verifies PLUS installation using: `python plus_toolkit_config.py`
2. Hand-Eye Calibration
 - a. NDI tracker connects.
 - b. Camera and stylus tracking data are collected.
 - c. The system computes and stores calibrated camera-to-tracker transformation.
3. Ultrasound Calibration
 - a. User prepares ultrasound for continuous capture.
 - b. Ultrasound frames and tracking transforms are acquired synchronously.
 - c. System generates a PLUS-compatible tracked sequence file: `.igs.mha`
4. Temporal Calibration
 - a. User runs: `python test_simulated_calibration.py --type temporal`
 - b. Delay value between ultrasound image stream and tracking stream is applied to align tracking and image data.
5. Spatial / Pivot Calibration
 - a. User runs:
`python test_simulated_calibration.py --type spatial`
`python test_simulated_calibration.py --type pivot`
 - b. Probe spatial relationship and pivot calibration parameters are saved for reconstruction.
6. 3D Volume Reconstruction
 - a. Using calibrated tracked sequence data, generates a 3D ultrasound volume.
 - b. User can view resulting `.mha` volumes using: `python view_mha_gui.py` or command-line viewer.

Deployment Environment:

- Local Windows 10/11 (64-bit)
- Python 3.8+
- PLUS ToolKit installed locally
- VTK + PySide6 for GUI visualization
- NDI tracker hardware connected for calibration